

Goals: Learn to use a class to set up an object type.

Do a simple example in detail.

Learn about instance variables, methods, constructors for objects and object types.

- An **object type** groups data and behavior.
- Every **object** has an **object type**
 - which is set up by writing a **class**, the name of the **class** is the name of the **type**.
 - **object data** is set up with **variables** = **instance variables**.
 - **behavior data** with **functions** = **methods**.

To plan any class = object type:

- 1) what **instance variables** needed? what data describes an **object**?
- 2) what **methods** apply to **objects**? what behavior?

Advice: To answer these write a simple piece of code to use **object**

Example 1 A counter object

```
Counter clicks = new Counter(); ← creates a new object
clicks.increment();
clicks.increment();
System.out.println(clicks.getValue()); //should say 2
```

② required methods

1. increment()
2. getValue()
3. constructor

① instance variables

1. int count

Coding to **object type**:

```
public class Counter {
    int count;
```

← class name matches type name

```
public Counter() {
    count = 0;
```

Constructor

1. it has the same name as the class.
2. no return type. not even void.

```
public void increment() {
    count++;
}
```

← coding a method is writing a function. should be public, not static.

```
public int getValue() {
    return count;
}
```

Rule: an object method isn't static.

Main Program Example

```
public static void main(String args[]) {
    Counter clicks = new Counter();
    clicks.increment();
    clicks.increment();
    System.out.println(clicks.getValue());
}
```

Example 2 A bank account

```
BankAccount myAccount = new BankAccount("Alex", 100);
myAccount.deposit(20.0);
myAccount.withdraw(50.0);
system.out.println(myAccount.getBalance());
```

① instance variables

1. string name
2. double balance

② required methods

1. constructor public BankAccount(string n, double b)
2. public void deposit(double d)
3. public void withdraw(double w)
4. public double getBalance()